

Gene Pulser® Electroprotocol

Cell Type Bacterial, gram positive

Molecules Electroporated DNA: pGT633, covalently closed circular form, a native 9.8kB erythromycin resistant *Lactobacillus* plasmid.

Species Used *Lactobacillus fermentum* , RF 14, a gastrointestinal isolate from pig intestine

Before the Pulse

Cell Growth Medium Lactobacilli MRS broth (Difco)

Growth Phase at Harvest O.D. (600) =0.8 (log phase cells)

Pre-pulse Incubation 0°C for 1 min

Wash Solution 3.5X SMEB; Luchansky *et.al.* (1988) Bio-Rad Bulletin **1350**:1-3

The Pulse

Instruments Used Gene Pulser® apparatus

Electroporation Temperature 0 °C

Electroporation Medium 3.5 x SMEB (1x = 272 mM sucrose, 1 mM MgCl₂)

Cuvette Gap 0.4 cm

Cell Density 10 (9) cells / ml

Voltage 2.5 kV

Volume of Cells 800 µl

Field Strength 6.25 kV/cm

DNA Concentration 10 µg

DNA Resuspension Buffer TE (10 mM Tris, 1 mM EDTA, pH 8.0)

Capacitor 25 µF

Volume of DNA 5 µl

Resistor (Pulse Controller) Not used**

After the Pulse

Time Constant 10 to 15 msec

Outgrowth Medium Lactobacilli MRS broth (Difco)
10 ml

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.

Ref:H.J. Connell, "Investigation of Methods for the Transformation of Gastrointestinal Strains of *Lactobacilli* with Plasmid pGT633" (1990).Ph.D. thesis, University of Otago, Dunedin, NEW ZEALAND (2) This work was carried out under the supervision of Dr. G. Tannock, Dept. of Micro., University of Otago,Box 56, Dunedin, NEW ZEALAND. PH: +64-3-4797713, Fax 64-3- 4741607. Questions regarding the availability of strains and the pGT633 should be directed to him.**It is NOT RECOMMENDED to use high voltage with out the Pulse Controller.

Outgrowth Temperature 37 °C

Length of Incubation 3 hours

Selection Method or Assay Used Erythromycin 25 µg / ml; requires a minimum expression time of 3 hr. to recover transformants

Electroporation Efficiency 5 x 10 (3) transformants / µg DNA, avg. Range: 4 X10 (2) to 4 X 10 (5)

Per Cent Survival 17 %

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Survey Number

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